RotSpot: Mobile-Based AI System for Freshness Detection of Fruits and Vegetables Using Convolutional Neural Networks

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Abstract— Food spoilage in fruits and vegetables remains a persistent issue affecting food quality, safety, and consumption efficiency, particularly in areas where manual freshness assessment is commonly practiced. This study presents RotSpot, an AI-based system for automated freshness detection of fruits and vegetables using image classification. The system employs a lightweight Convolutional Neural Network (CNN) based on the MobileNetV2 architecture with transfer learning to enable efficient and accurate classification of produce into fresh and rotten categories. A dataset consisting of ten types of fruits and vegetables, each classified into fresh and rotten categories (resulting in 20 classes), was used for model training, validation, and testing. To improve generalization, data augmentation techniques such as rotation, flipping, zooming, and contrast adjustments were applied. The trained model was deployed in a Flutter-based Android application using TensorFlow Lite, enabling offline real-time inference on mobile devices. Experimental results show that the model achieved a validation accuracy of 92.6%, while real-world field testing using locally sourced samples yielded an overall accuracy of 74%. Performance degradation under real-world conditions was attributed to variations in lighting, background complexity, camera quality, and differences in produce appearance compared to the training dataset. A detailed evaluation using precision, recall, and F1-score further highlighted challenges in distinguishing visually similar classes, particularly among rotten samples. Despite these limitations, the system demonstrated practical and functional classification capability in real-world environments. The results indicate that the proposed approach is a viable and efficient solution for mobile-based food freshness assessment, with potential applications in reducing food waste and improving consumer decision-making.

Keywords— Convolutional Neural Networks, MobileNetV2, transfer learning, image classification, food freshness detection, mobile application.

I. INTRODUCTION

Food spoilage remains a significant global issue, particularly for perishable goods such as fruits and vegetables. These food items are highly sensitive to environmental conditions and handling, leading to rapid deterioration in quality. In local contexts such as public markets in Tagbilaran City, Bohol, assessing freshness is commonly performed through manual inspection, which is subjective, inconsistent, and dependent on individual experience. This creates a need for accessible and reliable automated tools that can assist consumers, vendors, and households in making informed decisions regarding food quality.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have enabled the development of intelligent systems capable of analyzing visual data for classification tasks. In particular, Convolutional Neural Networks (CNNs) have demonstrated high effectiveness in image recognition due to their ability to automatically extract relevant features such as color, texture, and shape—key indicators of freshness and spoilage. Prior studies have validated the effectiveness of CNN-based approaches in food quality assessment. U. Amin et al. [1] developed a CNN-based system using transfer learning that achieved high accuracy in classifying fruits as fresh or rotten. Similarly, Y. Zhang et al. [2] proposed a multi-task CNN model capable of performing both classification and freshness detection, improving performance through shared feature learning. Furthermore, A. Sharma et al. [3] introduced a hybrid CNN-LSTM architecture to capture complex feature relationships and enhance classification accuracy. These studies highlight the capability of deep learning models in analyzing visual patterns associated with food quality. In addition, lightweight architectures such as MobileNetV2 have been widely adopted due to their efficiency and suitability for deployment on mobile and embedded devices [5].

Despite these advancements, a key limitation in existing studies is the heavy reliance on controlled or curated datasets, which may not accurately represent real-world conditions. Models trained on high-quality, well-lit, and

uniform images often experience performance degradation when exposed to variations in lighting, camera quality, background clutter, and object positioning. Furthermore, many implementations prioritize model accuracy without addressing real-world deployment constraints such as offline accessibility and usability for non-technical users.

To address these gaps, this study utilizes a dataset composed primarily of images sourced from publicly available platforms, including Kaggle datasets and stock image repositories, and evaluates the model using real-world images captured from fruits and vegetables purchased in local markets in Tagbilaran City, Bohol. This approach enables the assessment of how well models trained on internet-sourced datasets generalize to real-world conditions. The study focuses on ten (10) selected fruits and vegetables, each classified into fresh and rotten categories, resulting in a total of twenty (20) classes. A CNN-based model utilizing transfer learning with MobileNetV2 is developed and deployed as an Android application using TensorFlow Lite, enabling offline, real-time predictions without reliance on internet connectivity.

The developed application is designed to be simple and accessible, allowing users to capture or upload images and receive immediate freshness predictions along with confidence levels. This makes the system practical for everyday use by consumers, small-scale vendors, and households, particularly in environments with limited connectivity. The integration of a lightweight model with a mobile platform ensures portability, efficiency, and ease of use, even for non-technical users. However, the system is subject to several limitations. Model performance may be affected by variations in camera quality, lighting conditions, image angles, and background noise. Additionally, differences between the visual characteristics of produce in the training datasets and those found locally in Tagbilaran City may influence prediction accuracy. These factors highlight the importance of evaluating AI models in real-world environments beyond controlled datasets.

This study contributes a lightweight mobile-based Convolutional Neural Network (CNN) system for fruit and vegetable freshness detection, validated using both controlled datasets and real-world field testing in local market conditions. Unlike existing works that primarily evaluate performance under controlled environments, this study emphasizes practical deployment through an offline Android application and assesses model robustness under real-world variations such as lighting, background clutter, and image quality differences.

The study contributes to the field of intelligent food quality assessment by developing an efficient, lightweight, and offline-capable classification system while evaluating its robustness under real-world conditions. The findings support efforts in reducing food waste and improving food safety, aligning with Sustainable Development Goal 12 (Responsible Consumption and Production) and Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure).

II. MATERIALS AND METHODS

A. System Architecture

The proposed RotSpot system was developed using Python as the primary programming language due to its strong support for machine learning frameworks such as TensorFlow and Keras. Model training was performed on Google Colab, which provides GPU and CPU acceleration for efficient deep learning computation.

For image-based freshness detection, the MobileNetV2 convolutional neural network (CNN) architecture was selected as the backbone model. MobileNetV2 employs inverted residual blocks and linear bottlenecks, resulting in a lightweight yet efficient architecture suitable for mobile deployment and transfer learning [5]. It was selected due to its balance between accuracy and computational efficiency, making it ideal for real-time inference on resource-constrained mobile devices. Transfer learning was applied by initially freezing the pretrained backbone layers while training the classification head. The last layers were then fine-tuned to adapt the model specifically to the fruit and vegetable dataset. To improve generalization and robustness, a data augmentation pipeline was implemented using TensorFlow's Sequential API. Augmentation techniques included random horizontal flipping, rotation, zooming, contrast adjustment, and translation. These transformations simulate real-world variations in image capture conditions such as lighting changes, camera angles, and object positioning.

Input images were resized to 160 × 160 pixels and processed in RGB format. Before being fed into the network, images were normalized using the standard MobileNetV2 preprocessing function.

The model was trained in two stages:

Stage 1: Training the classification head for 20 epochs with frozen backbone layers

Stage 2: Fine-tuning the last 30 layers for 10 additional epochs using a reduced learning rate

A batch size of 32, Adam optimizer, and categorical cross-entropy loss function were used. Early stopping,

learning rate reduction, and model checkpointing were applied to prevent overfitting and retain the best-performing model.

This architecture enables RotSpot to perform efficient and accurate classification of fruit and vegetable freshness while remaining lightweight enough for mobile deployment.

B. Dataset Collection

The RotSpot dataset consists of ten selected fruits and vegetables, namely apple, banana, mango, orange, grapes, tomato, cucumber, carrot, potato, and bell pepper. Each fruit and vegetable is labeled into two categories: fresh and rotten, resulting in a total of 20 classes. The images used for training, validation, and testing were mainly collected from publicly available Kaggle datasets and iStock image repositories. Only clear images showing whole fruits and vegetables were included to maintain consistency across all classes. Each class contains approximately 500 images, which were divided into 70% for training, 15% for validation, and 15% for testing.

Table I: Dataset distribution of selected fruits and vegetables.

Class	Training	Validation	Testing	Real-World Testing
Apple – Fresh	350	75	75	15
Apple – Rotten	350	75	75	15
Banana – Fresh	350	75	75	15
Banana – Rotten	350	75	75	15
Mango – Fresh	350	75	75	15
Mango – Rotten	350	75	75	15
Orange – Fresh	350	75	75	15
Orange – Rotten	350	75	75	15
Grapes – Fresh	350	75	75	15
Grapes – Rotten	350	75	75	15
Tomato – Fresh	350	75	75	15
Tomato – Rotten	350	75	75	15
Cucumber – Fresh	350	75	75	15
Cucumber – Rotten	350	75	75	15
Carrot – Fresh	350	75	75	15
Carrot – Rotten	350	75	75	15
Potato – Fresh	350	75	75	15
Potato – Rotten	350	75	75	15
Bell Pepper – Fresh	350	75	75	15
Bell Pepper – Rotten	350	75	75	15

For real-world evaluation, 15 images per class were collected under indoor and outdoor conditions, resulting in approximately 300 total test images. Fruits and vegetables were grouped by type during testing to reduce inter-class confusion. For rotten samples, fresh produce was allowed to naturally decompose for approximately 12 days before imaging. The dataset includes variations in lighting, background, and orientation to improve model generalization when applied to real-world conditions.

C. Image Preprocessing and Augmentation

Prior to training, all images underwent preprocessing and augmentation to ensure consistency and improve model generalization.

1. Image Resizing and Normalization

All images were resized to 160 × 160 pixels, which matches the input requirement of MobileNetV2. Images were converted to RGB format and normalized to a [0, 1] range.

2. Data Augmentation

To simulate real-world conditions and reduce overfitting, the following augmentation techniques were applied:

- Random horizontal flipping
- Random rotation (±15%)
- Random zoom (±20%)
- Random contrast adjustment (±20%)
- Random translation (±10%)

These transformations help the model become robust to variations in lighting, orientation, and positioning.

3. Dataset Preparation

The dataset was loaded using TensorFlow's image_dataset_from_directory API, which automatically assigns labels and batches the data efficiently. Caching and prefetching were also used to optimize training performance.

Importantly, augmentation was applied only to the training dataset to avoid data leakage.

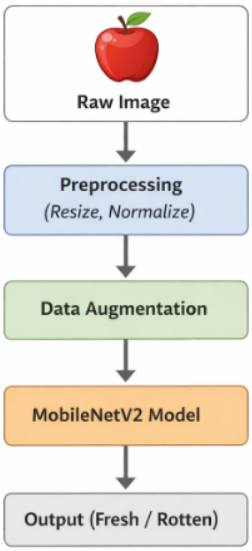

Fig. 1 Image Preprocessing and Data Augmentation Pipeline

D. Model Design and Training

The proposed system utilizes a Convolutional Neural Network (CNN) based on the MobileNetV2 architecture for image classification. MobileNetV2 was selected due to its lightweight structure and efficiency, making it suitable for deployment in mobile-based applications while maintaining high classification accuracy. The model was implemented using transfer learning, where the pretrained MobileNetV2 network (trained on ImageNet) was used as a feature extractor. The top classification layers were replaced with a custom fully connected layer consisting of a global average pooling layer, dropout regularization, and a dense output layer with softmax activation to support multi-class classification.

The system performs multi-class classification across 20 classes, representing fresh and rotten conditions for each of the ten selected fruits and vegetables. The output classes include categories such as fresh_apple, rotten_apple, fresh_banana, rotten_banana, and so on. This approach enables the model to not only determine freshness but also identify the specific type of produce. Training was conducted in two stages. In the first stage, the pretrained backbone of MobileNetV2 was frozen, and only the classification head was trained for 20 epochs. In the second stage, fine-tuning was performed by unfreezing the last layers of the network (approximately 30 layers) and training for an additional 10 epochs using a lower learning rate. This approach allows the model to retain general visual features while adapting to the specific dataset. The model was trained using a batch size of 32, with the Adam optimizer and categorical cross-entropy loss function. To improve training efficiency and prevent overfitting, several callbacks were utilized, including early stopping, learning rate reduction on plateau, and model checkpointing to save the best-performing model.

The trained model achieved a final validation accuracy of 92.6%, indicating strong performance in distinguishing between fresh and rotten produce across all classes. During inference, the model outputs a probability distribution across all classes, from which the class with the highest confidence score is selected as the final prediction. In the deployed Android application, only the top prediction and its corresponding confidence percentage are displayed to the user for simplicity and usability. For mobile deployment, the trained model was converted into TensorFlow Lite (.tflite) format using FP16 quantization, reducing model size and improving inference speed while maintaining acceptable accuracy. This enables efficient real-time classification on Android devices.

E. Application Development

The developed RotSpot system was deployed as an Android application to provide a user-friendly interface for real-time fruit and vegetable freshness detection. The application was built using the Flutter framework, enabling cross-platform development with a responsive and lightweight user interface. The application features a simple and intuitive design. Upon launching the app, users are presented with two options: selecting an image from the device gallery or capturing a new image using the device camera. Once an image is selected or captured, it is displayed on the screen and processed by the embedded AI model. For on-device inference, the trained CNN model was converted into TensorFlow Lite (.tflite) format using FP16 quantization, allowing efficient execution on mobile hardware. The application utilizes the TensorFlow Lite interpreter through the tflite_flutter package to load the model and perform predictions locally. This enables the system to operate offline, eliminating the need for an internet connection and ensuring faster response times.

During inference, the input image is resized to 160 × 160 pixels and converted into the appropriate numerical format before being passed to the model. The model outputs a probability distribution across all 20 classes, from which the class with the highest probability is selected as the final prediction.The application displays the predicted class label (e.g., fresh_apple or rotten_banana) along with the corresponding confidence percentage. This simplified output presentation ensures that users can easily interpret the results without requiring technical knowledge. Overall, the integration of the trained model into a mobile application demonstrates the practical applicability of the

proposed system, enabling real-time and accessible freshness detection for everyday use.

F. Testing and Validation

The performance of the proposed RotSpot system was evaluated through both dataset-based validation and real-world testing. During training, a portion of the dataset was reserved for validation to monitor model performance and prevent overfitting, achieving a final validation accuracy of 92.6%, which indicates strong classification capability across the defined 20 classes.

To assess the model's practical applicability, real-world testing was conducted using actual fruits and vegetables, with 15 images per class, resulting in approximately 300 test images across all fresh and rotten categories. The fruits and vegetables used in testing were sourced locally and evaluated in controlled groupings based on type to ensure consistency with the training data. To obtain rotten samples, fresh produce was allowed to naturally decompose over a period of approximately 12 days, from March 25 to April 6, prior to testing.

Images were captured using the developed Android application under varying conditions, including indoor and outdoor lighting, to simulate real-world usage scenarios. While efforts were made to maintain consistency, some images included minor background elements such as household objects, reflecting realistic environments.

The testing process involved capturing or uploading images through the mobile application, which performed on-device inference and displayed predictions along with confidence scores in real time. Sample images of the real-world testing setup and system usage are shown in Fig. 2.

Fig. 2 Real World Testing

The results showed that the system performs reliably under well-lit conditions and controlled framing. However, performance degradation was observed in cases involving poor lighting, complex backgrounds, and visually similar stages of spoilage. Additionally, variations in camera quality and image angles occasionally affected prediction confidence. Despite these limitations, the system demonstrated consistent performance and practical usability in real-world deployment scenarios.

III. RESULTS AND DISCUSSION

A. Model Performance

The proposed RotSpot system utilized a Convolutional Neural Network (CNN) based on the MobileNetV2 architecture, employing transfer learning with pre-trained ImageNet weights. The dataset consisted of approximately 7,001 training images and 1,500 validation images distributed across 20 classes, representing fresh and rotten conditions of selected fruits and vegetables. Training was conducted in two stages: initial training with frozen base layers followed by fine-tuning of the last layers to improve feature adaptation.

During the initial training phase, the model demonstrated rapid learning, with validation accuracy increasing significantly from 85.33% at Epoch 1 to over 91% within the first five epochs. This indicates that the pre-trained MobileNetV2 backbone effectively captured relevant visual features such as color and texture associated with freshness. As training progressed, both training and validation accuracy continued to improve steadily, reaching a peak validation accuracy of approximately 93.2%. The training accuracy achieved a maximum of approximately 94.3%, showing strong convergence of the model.

The fine-tuning phase further refined the model by unfreezing selected layers of the MobileNetV2 backbone and applying a lower learning rate. This stage stabilized the model's performance, resulting in a final validation accuracy of 92.6%. The slight difference between training and validation accuracy (approximately 1–2%) indicates minimal overfitting and demonstrates good generalization capability of the model on unseen data.

The training and validation loss curves exhibited a consistent decreasing trend, with validation loss stabilizing toward later epochs. This behavior suggests that the model effectively minimized classification errors while maintaining stability during training. No significant divergence between training and validation loss was observed, further confirming the robustness of the training process.

The performance trends of the model are illustrated in Fig. 3 and Fig. 4. Fig. 3 shows the training and validation accuracy curves, highlighting the steady improvement and

convergence of the model, while Fig. 4 presents the corresponding loss curves, demonstrating the reduction of error throughout the training process.

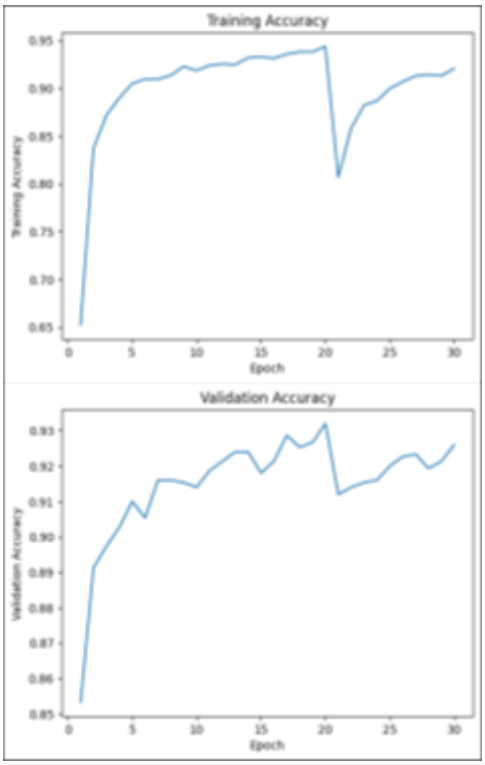

Fig. 3 Training and Validation Accuracy Curve

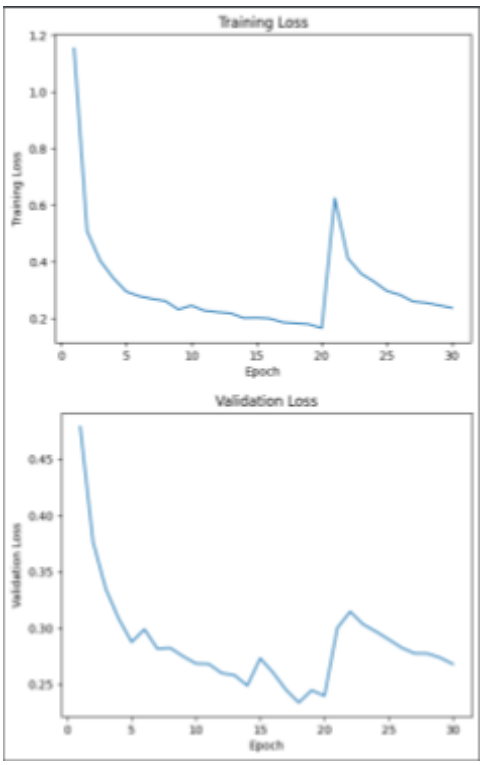

Fig. 4 Training and Validation Loss Curve

B. Field Testing Accuracy

The trained model was evaluated using real-world images consisting of 15 samples per class, resulting in a total of 300 test images across 20 classes. Based on the collected results, the system achieved an overall field-testing accuracy of approximately 74%, demonstrating its ability to generalize to real-world conditions beyond the training dataset.

Analysis of the results indicates that most fresh classes were classified with high accuracy, with several categories such as fresh cucumber and fresh tomato achieving near-perfect or perfect predictions. This suggests that the model effectively learned visual features such as color consistency and surface texture associated with fresh produce. However, performance on rotten classes exhibited greater variability. For instance, rotten apple samples were not correctly classified and were frequently misidentified as rotten mango and rotten potato. Similarly, rotten grape and rotten potato classes showed notable confusion with other rotten categories, likely due to similar visual characteristics such as discoloration, softening, and irregular texture patterns during spoilage.

Additionally, some misclassifications occurred between fresh classes, such as fresh mango being confused with fresh potato, and fresh bell pepper occasionally misclassified as fresh tomato or fresh carrot. These errors indicate that certain fruits and vegetables share overlapping visual features, particularly when captured under varying lighting conditions and backgrounds. Despite these challenges, the model maintained consistent performance across most categories and demonstrated practical usability for real-world freshness detection.

The detailed classification performance across all classes is illustrated in Fig.5, which presents the confusion matrix of the model based on real-world test data. The matrix highlights that most misclassifications occur among rotten classes, particularly between rotten mango, rotten potato, and rotten grape categories. In contrast, fresh classes exhibit stronger diagonal dominance, indicating higher classification accuracy and better feature distinction.

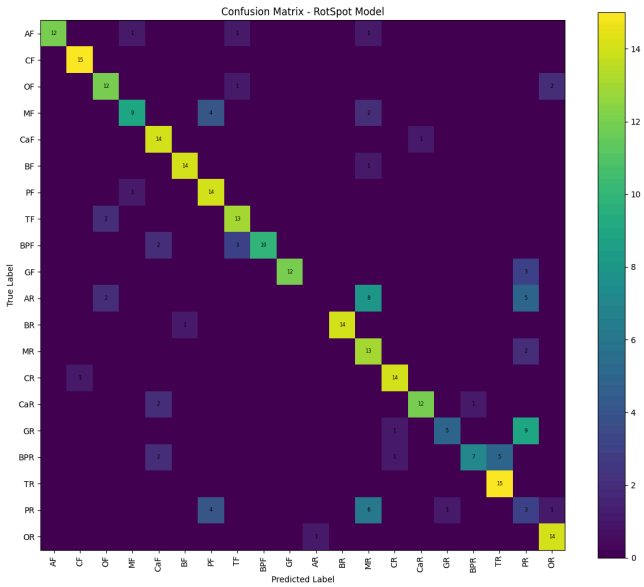

Fig. 5. Confusion matrix of the RotSpot model evaluated on real-world test data across 20 classes. Abbreviations: AF – Apple Fresh, CF – Cucumber Fresh, OF – Orange Fresh, MF – Mango Fresh, CaF – Carrot Fresh, BF – Banana Fresh, PF – Potato Fresh, TF – Tomato Fresh, BPF – Bell Pepper Fresh, GF – Grapes Fresh, AR – Apple Rotten, BR – Banana Rotten, MR – Mango Rotten, CR – Cucumber Rotten, CaR – Carrot Rotten, GR – Grapes Rotten, BPR – Bell Pepper Rotten, TR – Tomato Rotten, PR – Potato Rotten, OR – Orange Rotten.

The RotSpot system achieved a validation accuracy of 92.6% during training and a field-testing accuracy of 74%, indicating a noticeable performance gap between controlled and real-world conditions. This decrease in accuracy is expected in practical deployments, as real-world images introduce variations in lighting, camera quality, and background complexity that are not fully represented in the training dataset. Additionally, differences in the physical appearance of locally available fruits and vegetables, particularly those sourced in Bohol, may have contributed to classification inconsistencies, as these variations may differ from the images used during training.

Despite this gap, the system demonstrated strong performance in identifying most fresh produce and several rotten categories. The observed difficulty in classifying certain rotten classes can be attributed to visually similar characteristics such as discoloration, softening, and irregular texture, which increase the complexity of image-based classification tasks. Previous studies have shown that deep learning techniques are highly effective in agricultural image analysis, often achieving superior accuracy compared to traditional image processing methods [4]. Furthermore, deep learning-based approaches for fruit freshness detection, such as those proposed by Amin et al. [1] and Zhang et al. [2], report high accuracy under controlled datasets but acknowledge challenges when models are applied to real-world conditions with varying environmental factors.

Compared to existing deep learning models, including hybrid architectures such as CNN-LSTM [3], the RotSpot system demonstrates competitive performance during validation while emphasizing practical deployment through a mobile-based platform. Unlike many studies that primarily report dataset accuracy, this implementation extends its contribution by validating the model using real-world images, providing a more realistic evaluation of system performance. The use of the MobileNetV2 architecture [5] further supports efficient computation, enabling deployment on resource-constrained devices while maintaining high accuracy.

In comparison with existing mobile applications such as PlantNet and LeafSnap, which typically require internet connectivity for processing, the RotSpot system offers several advantages. The application operates entirely offline, eliminating dependency on network access and enabling usability in remote or low-connectivity environments. Additionally, the application is lightweight, with an approximate size of 60.54 MB, and features a simple and intuitive user interface, making it accessible even to non-technical users.

The system also contributes to broader societal goals, particularly Sustainable Development Goal 12 (Responsible Consumption and Production) by promoting awareness of food freshness and reducing unnecessary food waste. It further supports Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by introducing a scalable and accessible artificial intelligence solution that can be deployed on widely available mobile devices.

Overall, the results demonstrate that the RotSpot system is a viable and practical tool for fruit and vegetable freshness detection, with strong potential for real-world application. Future improvements may include expanding the dataset to incorporate more diverse and locally representative samples, as well as integrating preprocessing techniques such as background removal and lighting normalization to further enhance model robustness and accuracy.

C. Precision, Recall, and F1-Score Analysis

The performance of the proposed RotSpot system was further evaluated using precision, recall, and F1-score metrics to provide a more detailed assessment of classification performance across all classes. These metrics provide a deeper understanding of model behavior beyond

overall accuracy, particularly in multi-class classification tasks where class-wise performance varies significantly.

Precision measures the proportion of correctly predicted samples among all predictions made for a specific class, while recall measures the proportion of correctly identified samples out of all actual instances of that class. The F1-score represents the harmonic mean of precision and recall, providing a balanced evaluation of classification performance. These metrics are particularly important in this study due to the presence of visually similar classes and uneven classification performance across fresh and rotten categories.

Based on the generated classification report, the model achieved an overall accuracy of 74.00%, with a macro-average precision of 0.7604, macro-average recall of 0.7400, and macro-average F1-score of 0.7285. The weighted averages closely match the macro averages, indicating relatively balanced class distribution and consistent performance across most categories.

Class-wise analysis shows that several fresh produce categories such as Cucumber Fresh (CF), Banana Fresh (BF), Carrot Fresh (CaF), and Grapes Fresh (GF) achieved strong performance, with F1-scores above 0.85, demonstrating the model's ability to effectively learn discriminative features such as color consistency, shape, and texture associated with fresh produce. Similarly, some rotten classes such as Banana Rotten (BR), Tomato Rotten (TR), and Cucumber Rotten (CR) also achieved high recall and F1-scores, indicating reliable detection in clearer spoilage cases.

However, performance degradation is evident in certain challenging classes. Notably, Apple Rotten (AR) achieved a precision, recall, and F1-score of 0.0000, indicating complete misclassification of this class during field testing. This suggests that the model failed to learn distinguishing features for this category under real-world conditions. Additionally, Potato Rotten (PR) and Grapes Rotten (GR) exhibited low F1-scores of 0.1622 and 0.4762, respectively, primarily due to misclassification into visually similar rotten categories. These errors are attributed to overlapping visual characteristics such as discoloration, wrinkling, and texture degradation, which are common among multiple spoiled produce types.

A detailed per-class breakdown of precision, recall, and F1-score values is presented in Fig. 6, which shows the terminal output of the classification report generated from the evaluation script. This figure provides a complete visualization of model performance across all 20 classes and highlights the variation in classification effectiveness between different produce types.

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===== CLASSIFICATION REPORT =====

              precision    recall  f1-score   support

          AF     1.0000    0.8000    0.8889        15
          CF     0.9375    1.0000    0.9677        15
          OF     0.7500    0.8000    0.7742        15
          MF     0.8182    0.6000    0.6923        15
         CaF     0.7000    0.9333    0.8000        15
          BF     0.9333    0.9333    0.9333        15
          PF     0.6364    0.9333    0.7568        15
          TF     0.7222    0.8667    0.7879        15
         BPF     0.9091    0.6667    0.7692        15
          GF     1.0000    0.8000    0.8889        15
          AR     0.0000    0.0000    0.0000        15
          BR     1.0000    0.9333    0.9655        15
          MR     0.4194    0.8667    0.5652        15
          CR     0.8750    0.9333    0.9032        15
         CaR     0.9231    0.8000    0.8571        15
          GR     0.8333    0.3333    0.4762        15
         BPR     1.0000    0.4667    0.6364        15
          TR     0.6818    1.0000    0.8108        15
          PR     0.1364    0.2000    0.1622        15
          OR     0.9333    0.9333    0.9333        15

    accuracy                         0.7400       300
   macro avg     0.7604    0.7400    0.7285       300
weighted avg     0.7604    0.7400    0.7285       300
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Fig. 6 Classification report of the RotSpot system showing per-class precision, recall, F1-score, and support values obtained from real-world test evaluation.

To provide a comprehensive evaluation of the proposed system, three levels of performance metrics were used in this study. The validation accuracy of 92.6% represents the model's performance on the controlled dataset split during training. The field testing accuracy of 74% reflects real-world performance using locally sourced images captured under varying environmental conditions. Additionally, the classification report provides a more detailed evaluation based on precision, recall, and F1-score computed from real-world test data. These evaluation metrics collectively highlight the performance gap between controlled dataset conditions and real-world deployment scenarios.

D. Error Analysis

The performance results of the RotSpot system reveal several classification inconsistencies that are primarily attributed to visual similarity between classes, dataset limitations, and real-world image variability. While the model achieved strong performance in several fresh produce categories, specific failure cases provide important insights into its limitations and generalization behavior.

A significant performance issue is observed in the Rotten Apple (AR) class, which achieved 0.0 precision,

recall, and F1-score, indicating complete misclassification of all samples in real-world testing. This suggests that the model was unable to learn distinguishing visual features for rotten apple instances under real-world conditions. A likely cause of this failure is the limited visual variability of rotten apple samples in the dataset compared to other classes, resulting in poor feature generalization during inference.

In addition, confusion among rotten classes such as Rotten Mango (MR), Rotten Potato (PR), and Rotten Grape (GR) is frequently observed in the classification report and confusion matrix. These misclassifications are mainly due to the similarity in visual degradation patterns across different fruits and vegetables, including discoloration, texture softening, and shape deformation. Since these spoilage characteristics are not highly distinctive between classes, the model often struggles to assign correct labels, leading to cross-class confusion.

Conversely, fresh produce classes such as Cucumber Fresh (CF), Banana Fresh (BF), and Carrot Fresh (CaF) demonstrate significantly better performance, with higher precision and recall values. This improved performance can be attributed to the more consistent visual features of fresh produce, such as stable color distribution, clearer shape structure, and less variation compared to rotten samples. These features are more easily captured by the MobileNetV2 feature extractor, resulting in more reliable classification outcomes.

Another contributing factor to misclassification is the variation in real-world image conditions, including changes in lighting, background complexity, camera quality, and object positioning. Unlike the training dataset, which contains more controlled and curated images, real-world samples introduce noise and inconsistencies that affect feature extraction and classification confidence. This domain gap between training and testing environments is a known challenge in deep learning-based image classification systems.

Overall, the observed errors highlight the sensitivity of the model to intra-class similarity in spoiled produce and external environmental variations. Despite these limitations, the system still demonstrates reasonable generalization capability, particularly for fresh produce categories, and maintains acceptable performance under unconstrained real-world conditions.

IV. CONCLUSION AND RECOMMENDATION

The RotSpot system was successfully developed for the classification of fruit and vegetable freshness using a lightweight Convolutional Neural Network (CNN) architecture. By leveraging transfer learning with the MobileNetV2 model, the network was trained on a dataset comprising 20 classes representing fresh and rotten conditions of selected produce. The trained model achieved a validation accuracy of 92.6% and demonstrated effective learning of visual features associated with freshness and spoilage.

The deployed Android application was capable of performing real-time, offline predictions, achieving a field-testing accuracy of 74% based on 300 real-world samples. These results confirm the system's practicality and usability in real-world environments, particularly in scenarios with limited or no internet connectivity. The lightweight nature of the application, with an approximate size of 60.54 MB, combined with its simple and intuitive user interface, further enhances accessibility for both technical and non-technical users. Despite the system's overall effectiveness, certain limitations were identified during testing. Variations in lighting conditions, background complexity, and camera quality affected prediction accuracy in some cases. Additionally, the model showed difficulty in distinguishing between certain rotten classes due to similar visual characteristics of spoilage. Differences between locally available produce and training dataset images also contributed to classification inconsistencies.

For future work, it is recommended to expand and refine the dataset to include more diverse and locally representative images that reflect the actual appearance of fruits and vegetables found in Tagbilaran City, Bohol, in order to improve model generalization and real-world reliability. Incorporating advanced preprocessing techniques, such as background removal and lighting normalization, may further enhance robustness against environmental variations. In addition, future studies may explore the effectiveness of training models using large-scale, uncurated online datasets to evaluate the achievable performance without extensive manual data collection and labeling. Beyond software improvements, the proposed methodology may also be extended to hardware-based applications, such as automated sorting systems capable of distinguishing fresh and rotten produce in real time. Such systems could be implemented in markets or supply chains to improve operational efficiency, ensure product quality, and enhance customer satisfaction while maintaining business integrity. Furthermore, integrating additional features such as real-time camera guidance, batch image analysis, and optional cloud-based data logging could significantly extend the functionality and scalability of the RotSpot system as a practical solution for food freshness detection and waste reduction.

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